THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE FULL SYLLABUS PAPER SET 01

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. To test the effect of oil temperature on puffing, the dough circles should have: 1

(a) different sizes **(b)** different thickness
(c) the same thickness **(d)** different flour each time

2. Oil temperature is best measured with a: 1

(a) beam balance **(b)** thermometer **(c)** stopwatch **(d)** ruler

3. Rhizobium helps the soil by: 1

(a) fixing atmospheric nitrogen **(b)** making soil acidic
(c) adding sand **(d)** removing water

4. Which habit keeps us healthy? 1

(a) Regular exercise and adequate sleep **(b)** Using tobacco
(c) Eating only fried foods **(d)** Skipping meals

5. Chickenpox and measles are prevented by: 1

(a) vaccination and isolation **(b)** antibiotics
(c) cold water bath **(d)** vitamin pills

6. To reverse the poles of an electromagnet, we: 1

(a) add more turns **(b)** remove the core
(c) reverse the direction of current **(d)** use a thicker wire

7. A cell stops working when: 1

(a) the wire is straight **(b)** it is shaken
(c) chemicals get used up **(d)** it is warmed

8. The two kinds of static charge are: 1

(a) high and low **(b)** north and south
(c) positive and negative **(d)** hot and cold

9. A force of 240 N acts perpendicularly on an area of 4 m². The pressure is: 1

(a) 120 N/m² **(b)** 60 N/m² **(c)** 960 N/m² **(d)** 30 N/m²

10. Overhead water tanks are placed at a height to: 1

(a) reduce pressure **(b)** increase water pressure at taps
(c) make water warm **(d)** store more heat

11. Gases mix easily because their particles: 1

(a) are fixed **(b)** are closely packed
(c) are free and widely spaced **(d)** do not move

12. Rust is a: 1

(a) compound (iron oxide) **(b)** alloy of iron
(c) mixture of gases **(d)** pure element

13. Gold can be called a mineral and a metal because it is: 1

(a) an alloy
(b) only a compound
(c) a naturally occurring element that is a metal
(d) only a mixture

14. The density of water is about: 1

(a) 100 g/cm³ **(b)** 10 g/cm³ **(c)** 0.1 g/cm³ **(d)** 1 g/cm³

15. The Moon is visible because it: 1

(a) refracts only **(b)** produces its own light
(c) reflects sunlight **(d)** absorbs all sunlight

16. Vikram Sarabhai Space Centre is located in: 1

(a) Thiruvananthapuram **(b)** Sriharikota
(c) Hyderabad **(d)** Bengaluru

17. Organisms that eat both plants and animals are: 1

(a) carnivores **(b)** omnivores **(c)** producers **(d)** herbivores

18. Decomposers help the ecosystem by: 1

(a) making oxygen **(b)** hunting prey
(c) producing sunlight **(d)** recycling nutrients

19. The triple planetary crisis includes climate change, pollution and: 1

(a) rainfall **(b)** earthquakes
(c) biodiversity loss **(d)** lunar eclipse

20. The ozone layer protects life from: 1

(a) harmful ultraviolet rays **(b)** infrared only
(c) visible light **(d)** sound

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. What is biogas? Name its main gas. 2

22. What is antibiotic resistance? 2

23. Give two uses of rechargeable batteries. 2

24. Write two safety measures during lightning. 2

25. State two differences between mixtures and compounds. 2

26. Give one example each of a solid–liquid and liquid–liquid solution. 2

27. Mention two ecosystem services. 2

28. Name the triple planetary crisis. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Why should experiments be done safely? Give two precautions while frying. 3

30. Explain why weight differs on different planets while mass does not. 3

31. Why does a candle's wax change state near the flame? Describe. 3

32. Explain why light reflects in many directions from a rough wall. 3

33. Describe the activity of observing the Moon at the same time on successive days. 3

34. Explain how air, water and sunlight sustain life. 3

35. Why do phone batteries need charging more often after a few years? 3

OR

Compare the magnetic effect and heating effect of current with one use each. 3

36. Give three examples of ecosystem services in daily life. 3

OR

How are producers essential for other organisms? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. Write an essay on why prevention is better than cure, using at least five points. 5

OR

Describe the role of hygiene, clean water and sanitation in preventing communicable diseases with three examples. 5

38. Describe the destructive effects of cyclones and the precautions that can reduce loss. 5

OR

Explain why a rubber sucker sticks and why it becomes hard to remove, using the idea of pressure differences. 5

39. Explain saturated solution preparation and how to convert it into an unsaturated solution by two methods. 5

OR

Calculate the density of three liquids with the following (mass, volume): A (50 g, 50 mL), B (80 g, 100 mL), C (120 g, 100 mL). Which will float on water? 5

40. Explain how human activities affect ecosystems and what individuals can do. 5

OR

Write about Vrikshayurveda and traditional approaches to soil health compared with modern chemical farming. 5

END OF QUESTION PAPER — Total: 80 Marks

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